

Infant Mortality in the United States

A Descriptive Epidemiological Analysis, 2007–2023

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Data Source: CDC WONDER Linked Birth/Infant Deaths Database

Analysis Software: SAS Version 9.4

Date: May 23, 2026

Executive Summary

Infant mortality is defined by the CDC as the death of a live-born infant before the first birthday, and it is one of the most widely used indicators of population health and the overall quality of a nation's maternal and child health system (CDC, 2024). Despite decades of progress, the United States continues to report higher infant mortality rates than most peer high-income nations, with significant variation across racial, ethnic, and geographic groups (Ely & Driscoll, 2023).

This report presents a descriptive epidemiological analysis of U.S. infant mortality trends from 2007 to 2023 using publicly available data from the CDC WONDER Linked Birth/Infant Deaths database. The analysis examines national trends over time, racial and ethnic disparities, leading causes of infant death, and geographic variation across states and U.S. Census regions. All analyses were conducted using SAS Version 9.4.

Key Findings

1. The U.S. infant mortality rate declined over the 2007–2023 study period, reflecting sustained improvements in neonatal intensive care, prenatal services, and public health programming. A notable change in trend was observed around 2020–2021, coinciding with disruptions in healthcare access during the COVID-19 pandemic.
2. Substantial racial and ethnic disparities in infant mortality persisted throughout the study period. Non-Hispanic Black infants consistently experienced mortality rates approximately 2–3 times higher than Non-Hispanic White infants. While rates declined across all groups, the relative disparity remained largely unchanged over the 17-year period.
3. Congenital malformations, deformations and chromosomal abnormalities were the leading cause of infant death with 80,643 total deaths, followed by disorders related to short gestation and low birthweight with 65,846 deaths. Together these two causes accounted for approximately 44% of all rankable infant deaths from 2007 to 2023.
4. Southern states consistently reported the highest infant mortality rates. Mississippi had the highest average rate at 9.1 per 1,000 live births, nearly double the national average, followed by Alabama (8.2) and Louisiana (7.9). The South region maintained the highest regional infant mortality rate throughout the entire study period.

Methods

Data Source

Data was obtained from the CDC WONDER Linked Birth/Infant Deaths database for the years 2007 through 2023. This database links birth certificates with infant death certificates and is maintained by the National Center for Health Statistics (NCHS) within the Centers for Disease

Control and Prevention (CDC). The database is publicly available at wonder.cdc.gov. Three separate queries were constructed to obtain data by race/ethnicity and year, cause of death and year, and state and year.

Study Population

All live births and infant deaths occurring in the United States from January 1, 2007, through December 31, 2023, were included in this analysis. Infant mortality was defined as death before the first birthday, consistent with the standard epidemiological definition used by NCHS and the World Health Organization.

Measures

The infant mortality rate was calculated as the number of infant deaths per 1,000 live births. Rates flagged as unreliable by CDC WONDER indicating fewer than 20 deaths were excluded from trend analyses and interpreted with caution. Suppressed death counts indicating fewer than 10 deaths were coded as missing to protect individual privacy, consistent with CDC data use guidelines.

Race and ethnicity were classified using Mother's Bridged Race/Hispanic Origin to ensure consistency across the full 2007–2023 study period, consistent with CDC WONDER data availability for this time span. Four analytic race/ethnicity categories were used: Non-Hispanic White, Non-Hispanic Black, Hispanic, and Non-Hispanic Other.

Cause of death analysis was restricted to NCHS rankable causes of infant death as defined by the ICD-10 130 Cause List in CDC WONDER, denoted by the # symbol. Rankable causes are mutually exclusive and non-overlapping categories used in all official CDC infant mortality publications, ensuring that deaths are not doubly counted across categories.

States were grouped into four U.S. Census Bureau geographic regions: Northeast, Midwest, South, and West, consistent with standard federal geographic classification used in public health surveillance reporting.

Statistical Analysis

All data management and statistical analyses were conducted using SAS Version 9.4. Descriptive statistics including death counts, birth counts, mortality rates, and temporal trend analyses were performed using PROC MEANS and PROC FREQ. State-level average rates were calculated across the 2007–2023 study period excluding unreliable estimates. Data visualizations were produced using PROC SGPLOT.

Results

National Infant Mortality Trend, 2007–2023

Figure 1 presents the overall U.S. infant mortality rate from 2007 through 2023. The national rate demonstrated a general declining trend over the study period. A notable change in the trajectory of decline was observed in 2020–2021, coinciding with the onset of the COVID-19 pandemic. Disruptions to routine prenatal care, changes in hospital capacity, and the indirect health effects of the pandemic on pregnant women may have contributed to this pattern. The trend resumed its downward trajectory in subsequent years.

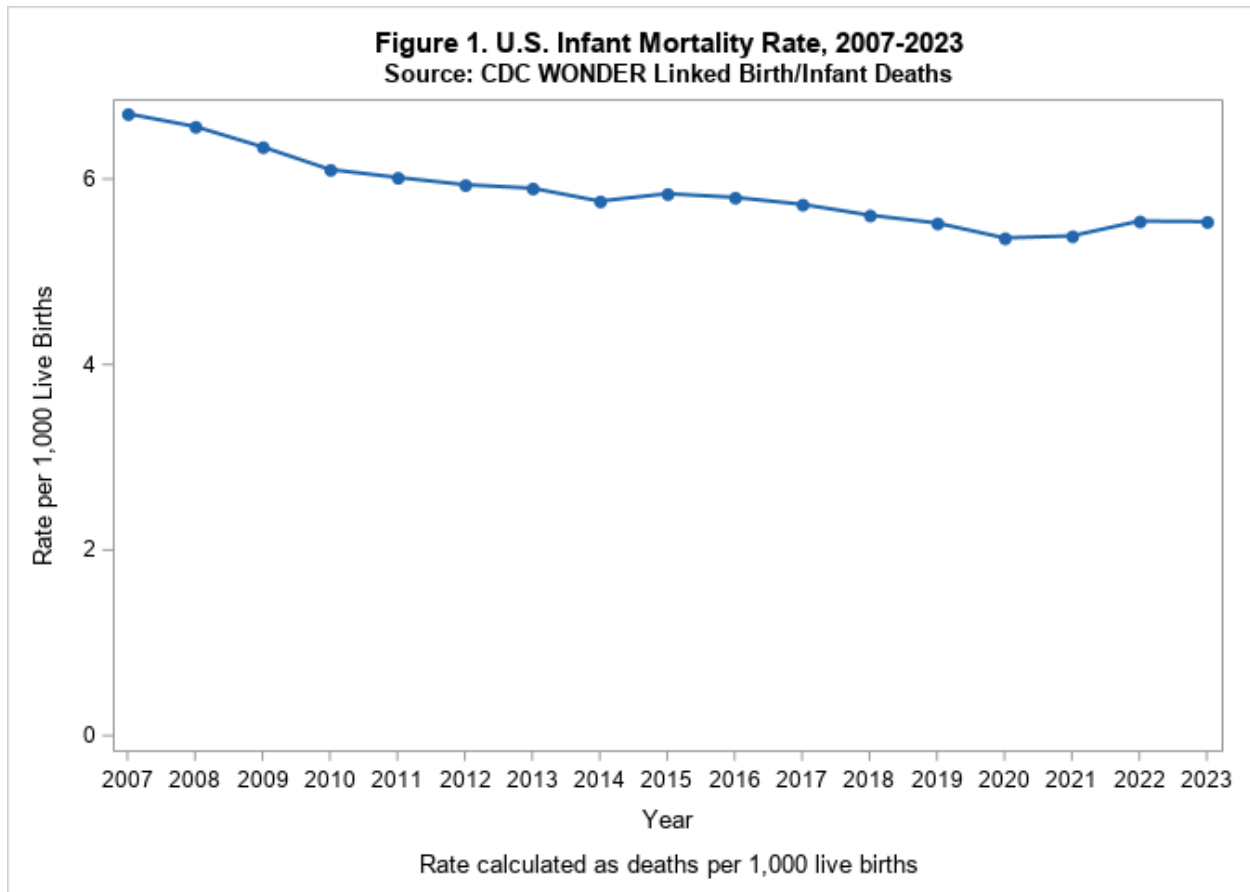


Figure 1. U.S. Infant Mortality Rate per 1,000 Live Births, 2007–2023. Source: CDC WONDER Linked Birth/Infant Deaths Database.

Racial and Ethnic Disparities in Infant Mortality, 2007–2023

Figure 2 illustrates infant mortality trends stratified by race and ethnicity from 2007 to 2023. Substantial and persistent racial and ethnic disparities in infant mortality were observed across all years of the study period. Non-Hispanic Black infants consistently experienced the highest

mortality rates, approximately 2–3 times higher than Non-Hispanic White infants throughout all 17 years analyzed. These persistent disparities likely reflect the compounding effects of structural racism, differential access to quality prenatal and obstetric care, higher prevalence of preterm birth among Black mothers, and the chronic stress associated with experiences of discrimination — factors that cannot be addressed through clinical interventions alone (Macdorman & Mathews, 2011) (Mathews & Driscoll, 2017).

While infant mortality rates declined across all racial and ethnic groups over the study period, the relative disparity between Non-Hispanic Black and Non-Hispanic White infants remained largely unchanged, indicating that overall improvements in infant survival have not translated into meaningful reductions in racial inequity. Hispanic infants demonstrated rates between those of Non-Hispanic White and Non-Hispanic Black infants, while Non-Hispanic Other race infants showed the most variable rates across the study period, likely reflecting the heterogeneous nature of this broadly defined category.

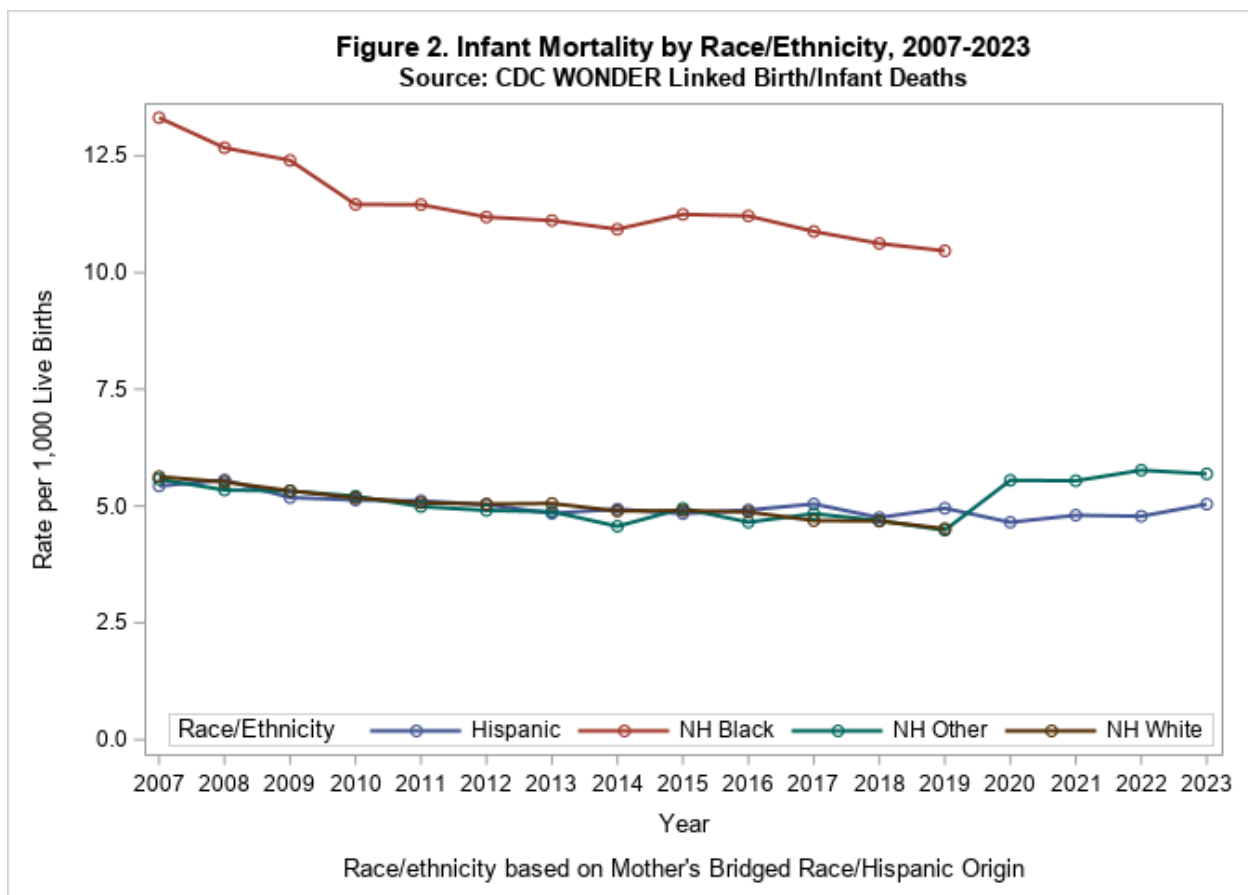


Figure 2. U.S. Infant Mortality Rate by Race/Ethnicity per 1,000 Live Births, 2007–2023. Source: CDC WONDER Linked Birth/Infant Deaths Database. Race/ethnicity based on Mother's Bridged Race/Hispanic Origin.

Leading Causes of Infant Death, 2007–2023

Figure 3 presents the top 10 leading causes of infant death in the United States from 2007 to 2023, based on NCHS rankable causes using the ICD-10 130 Cause List. Congenital malformations, deformations and chromosomal abnormalities (ICD-10: Q00–Q99) were the leading cause of infant death with 80,643 total deaths over the study period. Disorders related to short gestation and low birthweight (ICD-10: P07) ranked second with 65,846 deaths. Together these two causes accounted for approximately 44% of all rankable infant deaths from 2007 to 2023. Sudden infant death syndrome (SIDS, ICD-10: R95) ranked third with 28,611 deaths, followed by newborn complications related to maternal complications of pregnancy (P01) with 24,596 deaths, and accidents and unintentional injuries (V01–X59) with 20,901 deaths.

These findings highlight that most infant deaths in the United States are attributable to conditions originating in the perinatal period and congenital conditions, many of which are potentially preventable or amenable to medical intervention through improved preconception care, prenatal screening, and advances in neonatal medicine.

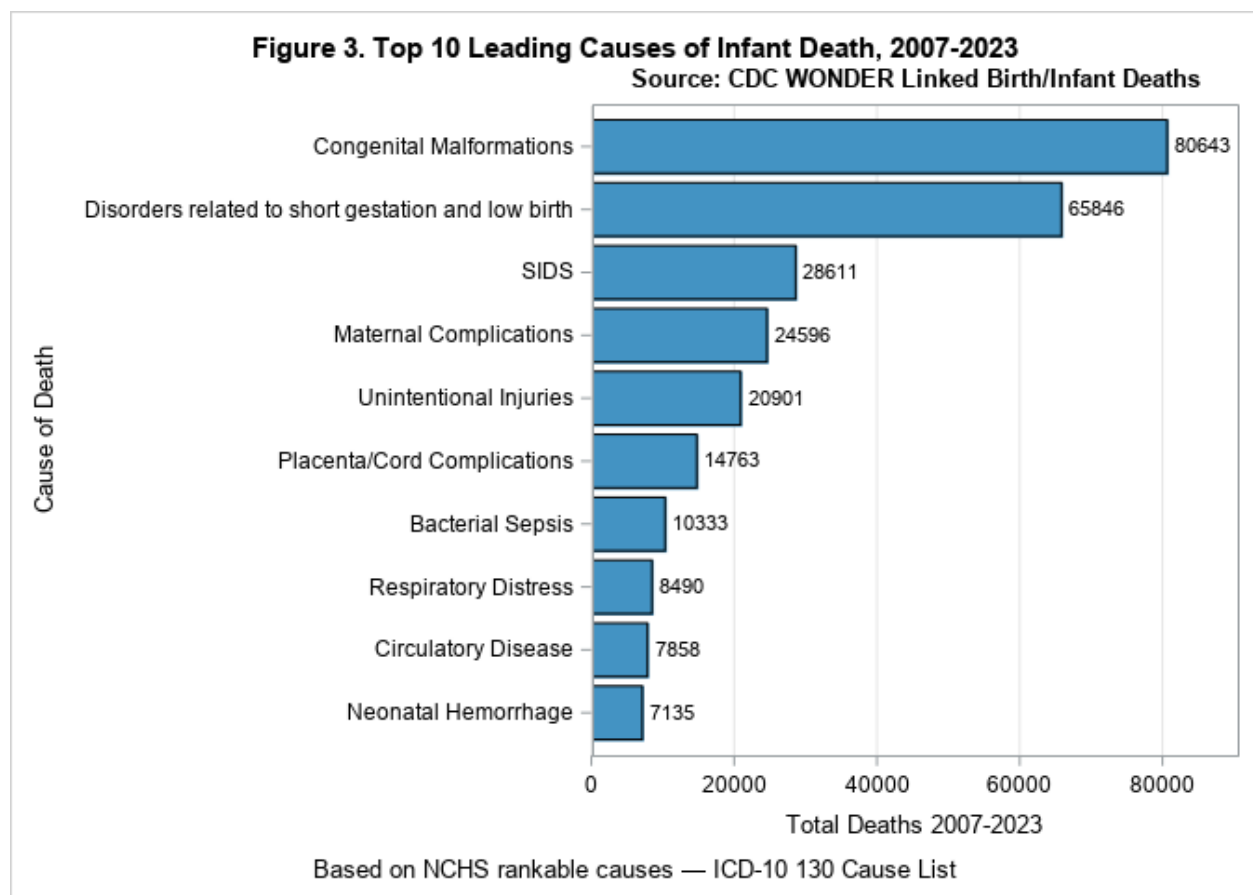


Figure 3. Top 10 Leading Causes of Infant Death, United States, 2007–2023. Source: CDC WONDER Linked Birth/Infant Deaths Database. Based on NCHS rankable causes — ICD-10 130 Cause List.

Geographic Variation in Infant Mortality, 2007–2023

Figures 4 and 5 illustrate geographic variation in infant mortality across U.S. states and Census regions (U.S. Census Bureau, 1984). Marked geographic disparities in infant mortality were observed, with Southern states consistently reporting the highest average rates over the 2007–2023 study period.

Mississippi had the highest average infant mortality rate at 9.1 per 1,000 live births, followed by Alabama (8.2), Louisiana (7.9), the District of Columbia (7.8), Arkansas (7.6), Oklahoma (7.3), Ohio (7.3), North Carolina (7.2), Tennessee (7.2), and South Carolina (7.1). Nine of the ten highest-burden states or jurisdictions were in the South, with Ohio being the only non-Southern state in the top 10. These states share common characteristics including higher rates of poverty, limited access to obstetric and neonatal care, higher prevalence of preterm birth, and larger proportions of non-Hispanic Black residents who face disproportionate risk of infant death.

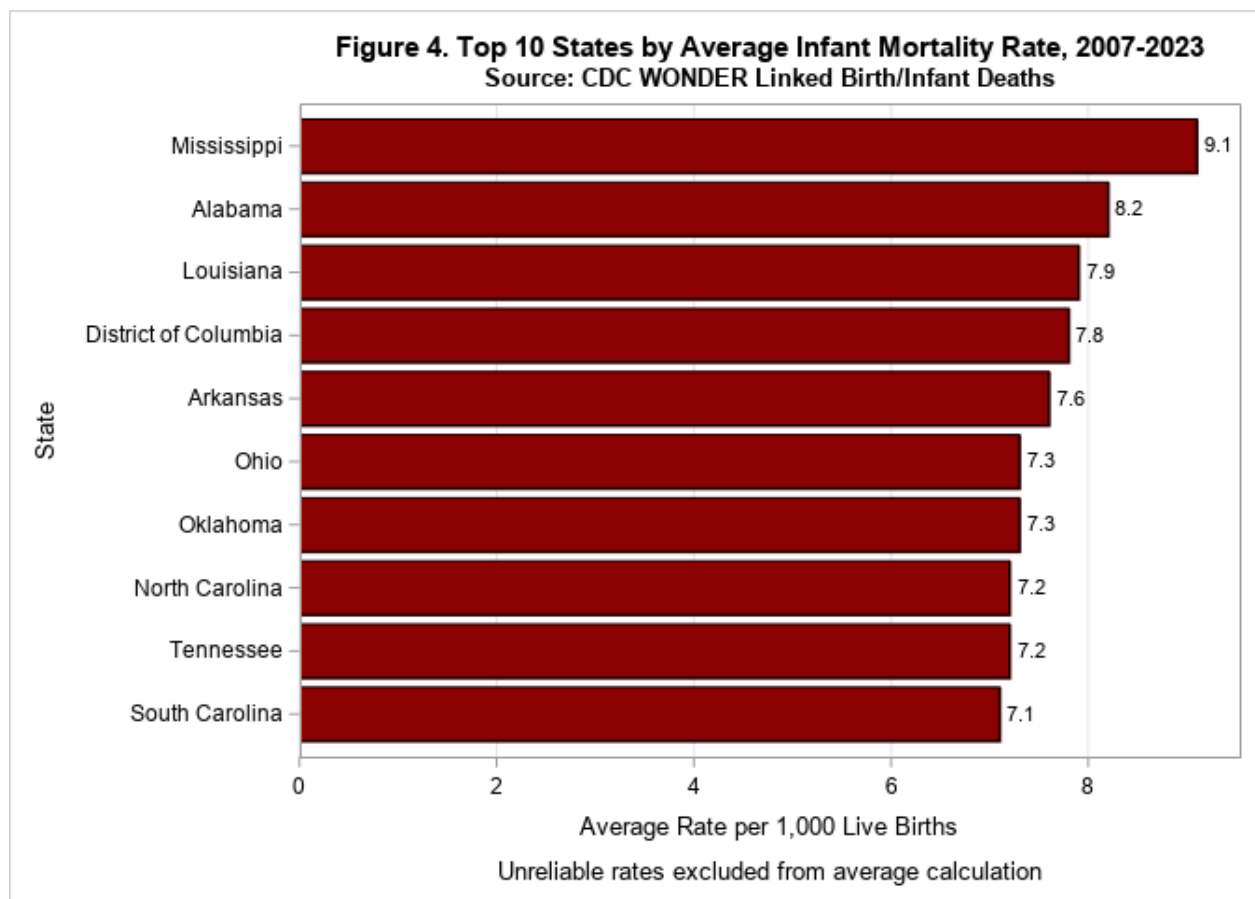


Figure 4. Top 10 States by Average Infant Mortality Rate per 1,000 Live Births, 2007–2023. Source: CDC WONDER. Unreliable rates excluded from average.

Regional trend analysis in Figure 5 confirmed that the South maintained the highest infant mortality rate throughout the entire 2007–2023 study period, while the Northeast consistently reported the lowest rates. The Midwest and West fell between these two extremes, with the Midwest showing rates closer to the South in some years. All four regions demonstrated declining trends over the study period, though the magnitude of decline varied.

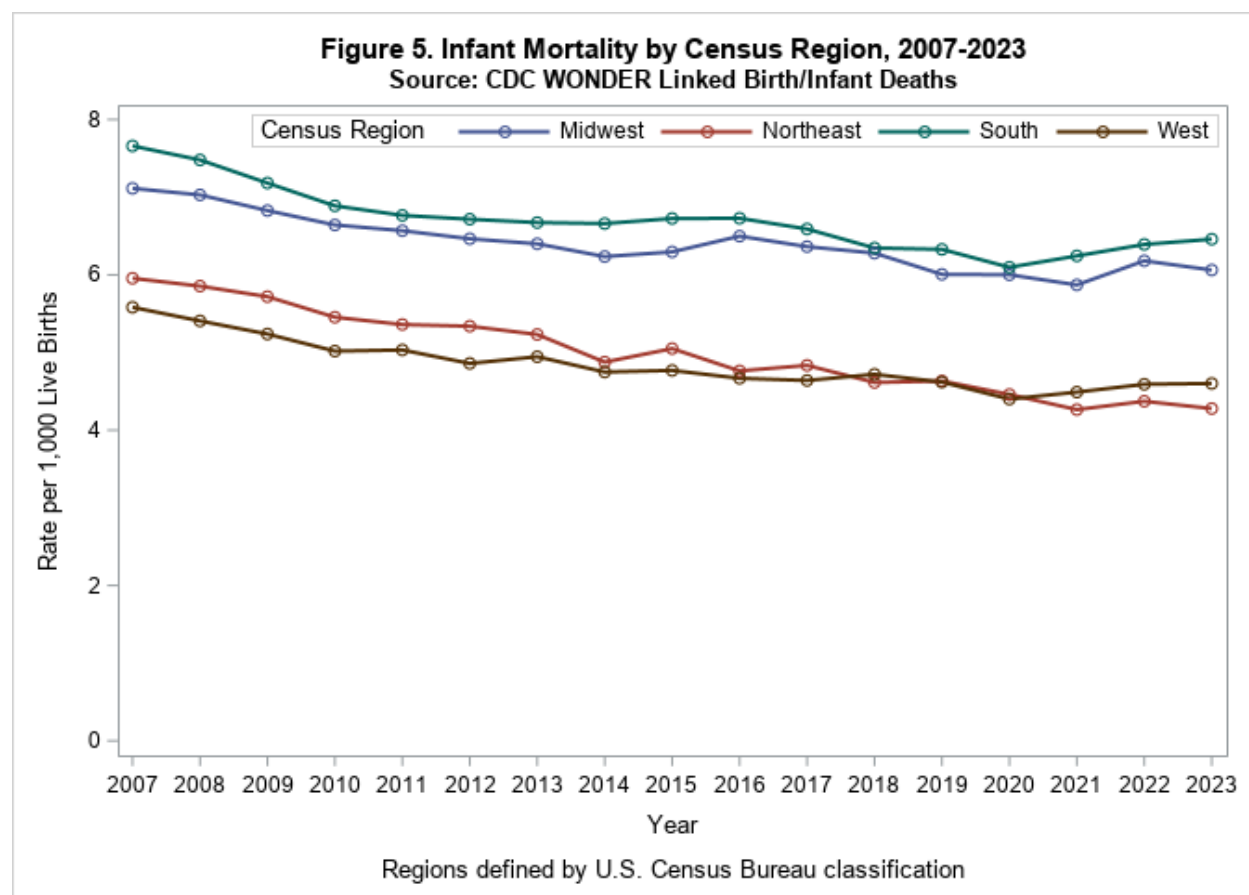


Figure 5. U.S. Infant Mortality Rate by Census Region per 1,000 Live Births, 2007–2023. Source: CDC WONDER. Regions per U.S. Census Bureau classification.

Conclusions

This analysis demonstrates that while U.S. infant mortality has declined over the 2007–2023 period, persistent and significant disparities remain across racial and ethnic groups and geographic regions. The gap between Non-Hispanic Black and Non-Hispanic White infant mortality rates has not narrowed meaningfully over 17 years of observation, indicating that broader systemic and structural factors continue to drive inequitable outcomes beyond what clinical improvements alone can address.

Southern states bear a disproportionate burden of infant mortality, highlighting the urgent need for targeted public health interventions, expanded access to prenatal and postpartum care, and policies addressing the social determinants of health including poverty, housing instability, food insecurity, and access to health insurance in these regions.

The two leading causes of infant death which are congenital malformations and short gestation/low birthweight, represent priority areas where continued investment in preconception care, genetic counseling, prenatal services, and neonatal intensive care could yield meaningful reductions in infant deaths. Addressing SIDS, the third leading cause, requires continued public health messaging around safe sleep practices, particularly in communities with the highest rates.

The findings of this analysis support continued investment in evidence-based maternal and child health programs, including Title V MCH Block Grant initiatives, the Maternal, Infant, and Early Childhood Home Visiting (MIECHV) program, and targeted interventions to reduce preterm birth and address racial disparities in birth outcomes (HRSA, 2023)

Limitations

1. Race and ethnicity were based on Mother's Bridged Race/Hispanic Origin to maintain consistency across study years, which may not fully capture individual self-identified race and ethnicity or reflect the experiences of multiracial individuals.
2. Rates based on fewer than 20 deaths were flagged as unreliable by CDC WONDER and interpreted with caution. Suppressed counts of fewer than 10 deaths were excluded from analysis, which may underrepresent mortality in small population subgroups.
3. This is a descriptive ecological analysis and cannot establish causal relationships between risk factors and infant mortality outcomes. Observed geographic and racial disparities likely reflect complex interactions between individual, community, and structural determinants of health.
4. State-level analysis reflects average rates across the full study period and may mask within-state variation by county, urban/rural status, or subpopulation group.
5. Changes in ICD-10 coding practices over the study period may affect the comparability of cause-of-death data across years, particularly for conditions such as SIDS where coding conventions have evolved.

References

- CDC. (2024, September 16). *Infant Mortality*. Maternal Infant Health.
<https://www.cdc.gov/maternal-infant-health/infant-mortality/index.html>
- Ely, D., & Driscoll, A. (2023). Infant Mortality in the United States, 2021: Data From the Period Linked Birth/Infant Death File. *National Vital Statistics Reports*, 72(11).
<https://www.cdc.gov/nchs/data/nvsr/nvsr72/nvsr72-11.pdf>
- HRSA. (2023, December). *Title V Maternal and Child Health (MCH) Services Block Grant | MCHB*. Mchb.hrsa.gov. <https://mchb.hrsa.gov/programs-impact/title-v-maternal-child-health-mch-services-block-grant>
- Macdorman, M. F., & Mathews, T. J. (2011). *Understanding Racial and Ethnic Disparities in U.S. Infant Mortality Rates*. <https://www.cdc.gov/nchs/data/databriefs/db74.pdf>
- Mathews, T., & Driscoll, A. (2017). *Trends in Infant Mortality in the United States, 2005-2014*. <https://www.cdc.gov/nchs/data/databriefs/db279.pdf>
- U.S. Census Bureau. (1984). *Census regions and divisions of the United States*.
https://www2.census.gov/geo/pdfs/maps-data/maps/reference/us_regdiv.pdf